

Python AI Developer Technical Test

Duration: 60 Minutes

Problem Statement

Build a simple **AI Marketing Automation System** using Python.

The system should be able to:

1. Generate marketing text from a given prompt.
2. Generate a marketing image URL from a given prompt.
3. Schedule marketing campaigns.
4. Send a marketing message to a specified phone number when the scheduled time arrives.

Campaign Details

Each campaign will contain:

- Campaign Name
- Prompt
- Phone Number
- Scheduled Date & Time

Example:

```
{  
  "campaign_name": "AI Course Launch",  
  "prompt": "Promote our new AI course",  
  "phone": "01913828774",  
  "schedule_time": "2026-06-15 10:00:00"  
}
```

Requirements

Implement a solution that:

- Creates and stores campaigns.
- Generates marketing text based on the campaign prompt.
- Generates an image URL based on the campaign prompt.
- Checks scheduled campaigns.
- Automatically sends a message to the provided phone number when the campaign time is reached.

Expected SMS Output

Sending marketing message to 01913828774

Campaign: AI Course Launch

Generated Text:

<generated marketing text>

Generated Image:

<generated image url>

Technical Requirements

- Use Object-Oriented Programming (OOP).
- Write clean, maintainable code.
- Include basic error handling.
- Organize the code properly.

Notes

- Real AI APIs are **not required**. You may use mock / Free API implementations for text and image generation.
- Real SMS sending is **not required**. Simulate the sending process using console output.
- Store campaigns in SQLite.
- Support multiple campaigns.
- Add logging.
- Expose the functionality through a simple REST API using FastAPI.

Submission: Source code + README with instructions to run the project.